

Determination of urinary lignans and phytoestrogen metabolites, potential antiestrogens and anticarcinogens, in urine of women on various habitual ...

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Recently two groups of compounds with diphenolic structure, the lignans and the isoflavonic phytoestrogens, were detected and identified in human urine and other biological fluids. These compounds are of great biological interest because they exhibit both in vitro and in vivo weak estrogenic and sometimes also antiestrogenic activities and many plant lignans have been shown to have anticarcinogenic, antiviral, antifungal and other interesting biological effects. The compounds found in relatively large amounts (10-1000 times more than estrogens) in urine are modified by intestinal bacteria from plant lignans and phytoestrogens, which are present in fiber-rich food such as grain and beans. They bind with low affinity to estrogen receptors and preliminary results suggest that they may induce production of sex hormone binding globulin (SHBG) in the liver and in this way may influence sex hormone metabolism and biological effects. Five compounds, the lignans enterolactone (Enl), enterodiols (End) and the isoflavonic phytoestrogen metabolites daidzein (Da), equol (Eq) and O-desmethylangolensin (O-Dma) were measured in urine by gas chromatography-mass spectrometry (selected ion monitoring) using deuterated internal standards in 5 groups of women (total number 53). The members of three dietary groups (omnivores, lactovegetarians and macrobiotics) were living in Boston and of two groups in Helsinki (omnivores and lactovegetarians). Until now measurements have been carried out in 94 72-h samples. The highest mean excretion of the most abundant compound, enterolactone, was found in the macrobiotic group and the lowest in the omnivore groups. Total mean 24-h excretion of enterolactone was 17,680 nmol in the macrobiotics, 4,170 nmol in the Boston lactovegetarians, 3,650 nmol in the Helsinki lactovegetarians, 2,460 nmol in the Helsinki omnivores and 2,050 nmol in the Boston omnivores. The other diphenols followed approximately the same pattern. In an earlier study the lowest excretion of enterolactone (1,040 nmol/24 h) was found in a group of postmenopausal apparently healthy breast cancer patients living in Boston. It is concluded that further studies are necessary to elucidate the possible role of these compounds in cancer and other diseases. However, the evidence obtained until now seems to justify the conclusion that these compounds may be among the dietary factors affording protection against hormone-dependent cancers in vegetarians and semivegetarians.